Chi (Faye) Zhang

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Education

The City University of New York (CUNY), Graduate Center — New York, NY
Ph.D. student in Computer Science | GPA: 3.73/4.0 — August 2024 – Present

Shandong University — Jinan, China
B.E. in Software Engineering | GPA:3.70/4.0 — September 2019 – June 2023

Research Interests

Security and privacy of agentic AI systems; accountability and digital forensics for AI agents; cloud audit and logging measurement; usable security; LLM-assisted education

Honors and Awards

- ACM CCS Individualized Cybersecurity Research Mentoring (iMentor) Workshop, Cohort — 2023
- Outstanding Student Award, School of Software, Shandong University — 2022
- Innovation and Entrepreneurship Scholarship, Shandong University — 2020, 2021, 2022

Selected Publications

- **Zhang, C.**, Ji, P. Toward Responsible Agentic AI Systems: Measuring the Audit Evidence Behind Agent Actions in the Cloud. Under review.
- **Zhang, C.**, Ji, P., Zhang, Z. (2026). "We Have Never Been Attacked": Unpacking the Illusion of Cybersecurity in Small Business. *Symposium on Usable Privacy and Security (SOUPS 2026) Poster*.
- Hanon, J., **Zhang, C.**, Ji, P. (2026). Privacy Tradeoffs in the Digital Era: Public Views on Data Use, Monitoring, and Accountability. *Symposium on Usable Privacy and Security (SOUPS 2026) Poster*.
- **Zhang, C.**, Liu, Y., Chen, X., Ji, P. (2026). What Keeps Agent Skills from Being Reusable? Evidence from 138K SKILL.md Files. *First Workshop on Agent Skills (AgentSkills), ACM CAIS 2026*.
- Chen, X., **Zhang, C.**, Ji, P., Liu, Y. (2026). SkillsMetric: Mapping the Detection Boundary of Static Analysis for Malicious Agent Skills. *First Workshop on Agent Skills (AgentSkills), ACM CAIS 2026*.
- Cai, J., **Zhang, C.**, Ji, P. (2025). InsightFlow: A Generative AI Approach to Streamlining Knowledge and Learning Paths. *International Conference on Web-Based Learning (ICWL 2025)*.
- Baravkar, S., **Zhang, C.**, Hassan, F., Cheng, L., Song, Z. (2024). Decoding and Answering Developers' Questions About Web Services Managed by Marketplaces. *IEEE International Conference on Software Services Engineering (SSE 2024)*, 194–203.

Research Experience

Accountability of Agentic AI Systems in the Cloud — Sep 2026 – Present
Supervisor: Prof. Ping Ji | Paper under review — New York, NY

- Conducted a large-scale measurement of audit evidence for LLM agent actions on Google Cloud and AWS, combining controlled agent deployments with scans of public agent and infrastructure-as-code repositories.
- Found that even with full logging, provider records attribute agent actions to the credential rather than the agent and cannot link an action to the model decision behind it; few public projects enable even the records that exist.
- Designed a tamper-evident agent-side record that links each action to its decision through identifiers that provider logs also carry, and validated it against provider records on both clouds.

Usable Security for Small Businesses — Oct 2024 – Present
Student Principal Investigator | Supervisor: Prof. Ping Ji | Full paper under review — New York, NY

- Designed and led a mixed-methods study of small-business cybersecurity ($N = 128$: interviews, surveys, and 102 founder intercepts), conducting all data collection, qualitative analysis, and writing.

- Found that owners often report never being attacked even though 81% of small businesses reported a breach in the previous year (ITRC 2025), and traced this **security illusion** to semantic narrowing, structural invisibility, and delegated trust.
- Derived design implications such as default-on audit signals and clear cloud responsibility boundaries.

OPAL: Open Platform for Adaptive Learning Analytics Sep 2026 – Present

PI: Prof. Ping Ji | Designer and developer New York, NY

- Designed and built **OPAL**, a course platform in which teachers upload their own materials and an LLM drafts quizzes for teacher approval; in use at Hunter College and several secondary schools.
- Built a Claude (Vertex AI) pipeline that extracts weekly knowledge points from course PDFs and generates questions that each cite their source sentence and explain the answer, plus a self-paced practice pool tracking per-concept mastery.
- Engineered the full stack (Next.js, TypeScript, PostgreSQL, Cloud Run) with strict per-course isolation and an append-only student action log for learning analytics.

Reusability and Security of Agent Skills Mar 2026 – May 2026

Self-initiated | Supervisor: Prof. Ping Ji | Two workshop papers (ACM CAIS 2026) New York, NY

- Measured 138,133 public SKILL.md files from 20,556 repositories against a defect taxonomy grounded in the official specification, finding that 91.8% contain at least one defect, mostly packaging problems such as weak routing metadata (first author).
- Showed with a routing stress test over 20,000 skills that these defects hurt retrieval, and proposed a quality-assured generation workflow (spec-aware prompting, linting, repair, safety gating).
- Co-developed **SkillsMetric**, a static analysis framework for malicious skills (AUC 0.93 across 16 attack types), and identified blind spots: host-destruction attacks (0% detected) and prompt injection (42%).

LLM Agents for Cyber Defense Nov 2025 – Feb 2026

Supervisor: Prof. Ping Ji | Poster (AIX Summit 2026) New York, NY

- Built and configured multi-agent experimental environments based on CybORG CAGE Challenges, conducted prompt ablation experiments and data analysis benchmarking GPT-4o and GPT-5.2 as autonomous Blue defenders.

AeDA Platform Development Sep 2024 – Present

Supervisor: Prof. Ping Ji New York, NY

- Full-stack developer for **AeDA** (Attainable eDucation for All), an AI-powered platform for personalized learning paths; built the mobile app and web portal and architected its data storage on GCP.

Knowledge Tracing for Personalized Learning Jul 2025 – Dec 2025

Supervisor: Prof. Ping Ji | Published (ICWL 2025) New York, NY

- Curated datasets and co-wrote the manuscript for a graph-aware knowledge tracing framework that combines educator-curated knowledge graphs with Transformer models in **AeDA**.

Research Assistant, University of Michigan Aug 2023 – Dec 2023

Web Services Research Ann Arbor and Dearborn, MI

- Classified developer questions about marketplace-managed web services to improve API documentation (IEEE SSE 2024).

Research Assistant, Shandong University Nov 2020 – Jun 2023

Supervisors: Prof. Lizhen Cui, Prof. Yonghui Xu Jinan, CN

- Co-authored a survey on music intervention (IJCS 2023) and built a rhythm-matching algorithm (ICCSE '21).
- Digitized Traditional Chinese Medicine manuscripts via image preprocessing, OCR, and data annotation.

Teaching

- CSCI 49395 – Big Data Analytics and Applied Privacy, Hunter College, CUNY (Teaching Assistant) Fall 2026
- CSCI 49392 – Advanced Operating Systems, Hunter College, CUNY (Teaching Assistant; independently delivered lectures) Spring 2026
- CSCI 127 – Introduction to Computer Science, Hunter College, CUNY (Teaching Assistant) Fall 2025

Professional Service

Reviewer

- EMNLP 2026 Industry Track 2026
- ICLR 2026 Workshop on Memory in Agentic AI (MemAgents) 2026
- First Workshop on Agent Skills (AgentSkills), ACM CAIS 2026 2026
- AAAI 2025 Undergraduate Consortium (UC) 2025

Skills

• Programming	Python, Java, C/C++, JavaScript, SQL, Bash, LaTeX.
• Frameworks & Tools	LangChain, CybORG, Wireshark, Kali Linux, Git, Docker, GCP.
• Security	Network Defense, LLM-based Cyber Defense, Threat Analysis.
• AI & Productivity	Prompt Engineering, Advanced AI Coding Agents (Claude Code, Codex), Skills.
• Research Methods	Mixed-Methods Study Design, Qualitative Analysis, AI-Augmented Workflow Design.